

Project Initialization and Planning Phase

Date	04 July 2026
Team ID	
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed Human Development Index (HDI) Prediction System uses Machine Learning to estimate a country's Human Development Index based on important socio-economic indicators. The project aims to simplify HDI prediction, improve accuracy, and provide users with a fast and user-friendly web application for research, education, and policy analysis.

Project Overview	
Objective	The primary objective of this project is to develop a Machine Learning-based HDI Prediction System that accurately predicts the Human Development Index using key indicators such as Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita. The system provides quick predictions through a simple Flask web application.
Scope	The project focuses on collecting and preprocessing the HDI dataset, training a Linear Regression model, and deploying it using Flask . It enables users to enter development indicators and instantly obtain the predicted HDI value along with the corresponding development category.
Problem Statement	
Description	Manual analysis and prediction of the Human Development Index require evaluating multiple socio-economic factors, making the

	process time-consuming and complex. Existing methods are not always user-friendly for students, researchers, and policymakers.
Impact	An automated HDI prediction system reduces manual effort, improves prediction accuracy, and supports faster decision-making in education, research, and policy planning.
Proposed Solution	
Approach	The solution uses a Linear Regression Machine Learning model trained on the Human Development Index dataset. The trained model is integrated with a Flask web application that accepts user inputs, predicts the HDI value, and displays the corresponding human development category.
Key Features	• Machine Learning-based HDI prediction using Linear Regression. • User-friendly web interface developed with Flask. • Instant prediction of HDI values. • Classification into Low, Medium, High, and Very High Human Development. • Simple and responsive interface for easy interaction.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5 or above
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	256 GB SSD or above
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn, pickle
Development Environment	IDE	Jupyter Notebook, Visual Studio Code
Data		
Data	Source, size, format	Kaggle dataset, csv, Approximately 195 country records with socio-economic indicators